Xiyan Zhou

Ann Arbor, Michigan, U.S., 48105 xiyanzho@umich.edu https://xiyanzho.github.io

EDUCATION

University of Michigan, Ann Arbor Ann Arbor, MI
B.S. Computer Science & B.S. Mathematics (Double Major) *Expected Graduation: December 2026*
GPA: 3.8 / 4.0

Coursework: Measure Theory, Convex Optimization, Bayesian Machine Learning, Numerical Linear Algebra, Real Analysis, Probability Theory, Machine Learning, Operating Systems, Data Structures & Algorithms, Combinatorics, Discrete Mathematics

SKILLS

Programming Languages: Python, C++, C, Java, SQL, MATLAB, JavaScript
Frameworks & Tools: JAX, Flax, PyTorch, NumPy, scikit-learn, LaTeX, Git, Bash, Linux
ML & Math: Reinforcement Learning, Bayesian Methods, Convex Optimization, CNNs, Statistical Modeling
Languages: Mandarin (native), English (proficient)

RESEARCH

Goal-Conditioned Contrastive Reinforcement Learning in Discrete Action Spaces

Research Assistant — advised by Kellen Kanarios *Jan 2026 – Present*

- Extended a JAX contrastive RL codebase (written by Kellen Kanarios, with a Gumbel-Softmax actor) for discrete actions: diagnosed and fixed action-encoding and observation-indexing bugs that had held navigation success at 3% (85% after the fix), and redesigned the actor objective. Built grid-world benchmarks (navigation, foraging, key–door) and goal encodings in which goal structure is a controlled variable, plus a matched goal-conditioned Double-DQN + HER baseline sharing environments, buffer, and evaluation.
- Characterized a failure mode of goal-conditioned RL: critics discriminate less sharply between actions when a goal admits multiple satisfying states. Success falls monotonically with satisfier count (key–door: 0.80 / 0.58 / 0.28 for 1 / 2 / 3 doors, non-overlapping seed ranges); a positional-goal control isolates the effect from board clutter; the Double-DQN baseline also degrades ($\times$0.64 vs. $\times$0.88 for contrastive RL), so the effect is a property of goal-conditioned value learning rather than of the contrastive objective.
- Designed *critic lift*, a probe of whether the critic's greedy action moves toward the correct target; it separates one- from two-satisfier runs across 15 runs without tracking task difficulty, whereas the contrastive training metric was decoupled from, and at times inverted relative to, control quality.
- A configuration screen showed the default InfoNCE direction was the worst of three (0.34 $\rightarrow$ 0.82 on the headline task); re-ran all experiments ($\sim$120 runs, $\geq$3 seeds per claim), withdrawing three earlier claims while the core effect held. Preprint in preparation.

AI-Driven Human Digital Twin Study (IOE SURE Program)

Undergraduate Research Assistant — advised by Prof. Jingwen Hu (UMTRI) *May 2026 – Present*

- Building the lower-extremity muscle segmentation component of UMTRI's HERMES human body model. Preprocessed 45 clinical CT scans into ML-ready left/right leg volumes with an automated pipeline (cropping, sagittal mirroring, boundary-overlap resolution) and a hand-labeled 3D ground-truth set for nnU-Net training.
- Trained and validated nnU-Net segmentation models achieving 0.92 mean Dice on the posterior compartment and 0.84 on the medial compartment, with posterior cross-sectional area error of 2.3–2.9% against hand-labeled ground truth.
- A model trained only on female scans reached 0.90 Dice on a held-out male case, comparable to its female performance (0.92) — an initial data point on unified vs. sex-specific HERMES parameterization.

PROJECTS

Diagnostic Study of Sparsity-Promoting ADMM for Multi-Obstacle Trajectory Optimization

Independent course research project — ECE 559: Optimization for Signal Processing & ML *Winter 2026*

- Formulated an $\ell_{2,1}$ group-sparsity trajectory optimization problem for a 2D double-integrator navigation task with non-convex obstacle-avoidance constraints, and derived a two-block ADMM splitting with over-relaxation and adaptive (residual-balancing) penalty scaling; achieved about 60% zero-input timesteps at lower control effort than quadratic MPC.
- Diagnosed a structural failure mode: the obstacle projection on the auxiliary variable Z does not transfer safety to the dynamically feasible trajectory X — across 10 randomized environments Z was collision-free in 8/10 and X in 0/10, and λ and iteration-budget (50–1000) sweeps showed the gap does not close with more iterations.

Max-Value Entropy Search vs. Expected Improvement for Bayesian Optimization

Course project — EECS 498: Bayesian Machine Learning *Winter 2026*

- Implemented Gaussian-process Bayesian optimization from scratch (marginal-likelihood hyperparameter fitting, BIC-based kernel selection, Expected Improvement), then re-implemented Max-Value Entropy Search (Wang & Jegelka, ICML 2017).
- Found that EI significantly outperformed my MES implementation in a paired comparison over 20 runs ($p = 1.32 \times 10^{-3}$); traced the likely cause to a small fitted GP lengthscale that made the objective resemble noise, undermining MES's reliance on estimating the distribution of the optimum.

Replicating and Extending SimGRL: Visual RL Robustness under Scene Perturbations

Team course project — EECS 498: Machine Learning Research Experience *Fall 2025*

- Replicated SimGRL (NeurIPS 2024) on DMControl-GB under a 60K-step budget on the Great Lakes HPC cluster; SimGRL beat the SVEA baseline in every evaluation mode, matching the original paper's trends.
- Designed, implemented, and tuned Hybrid-SRO (per-frame photometric perturbations on SimGRL's background overlays), one of four team extensions; it did not beat SimGRL on average but cut across-seed standard deviation from about 75 to about 24 while staying within 2% on Video Hard.

AWARDS & TEACHING

- **James B. Angell Scholar** — University of Michigan (March 2025); all A-range grades for two or more consecutive terms
- **University Honors** — University of Michigan (×4: Fall 2023, Winter 2024, Fall 2024, Winter 2025)
- **Grader** — MATH 571 (Graduate Numerical Linear Algebra), MATH 371 (Numerical Methods), EECS 203 (Discrete Mathematics)
- **Peer Tutor**, Math Learning Center (4 semesters)